When Robots Feel Gentle: Co-Designing Tactile Interfaces for Social Robots in Older Adult Care

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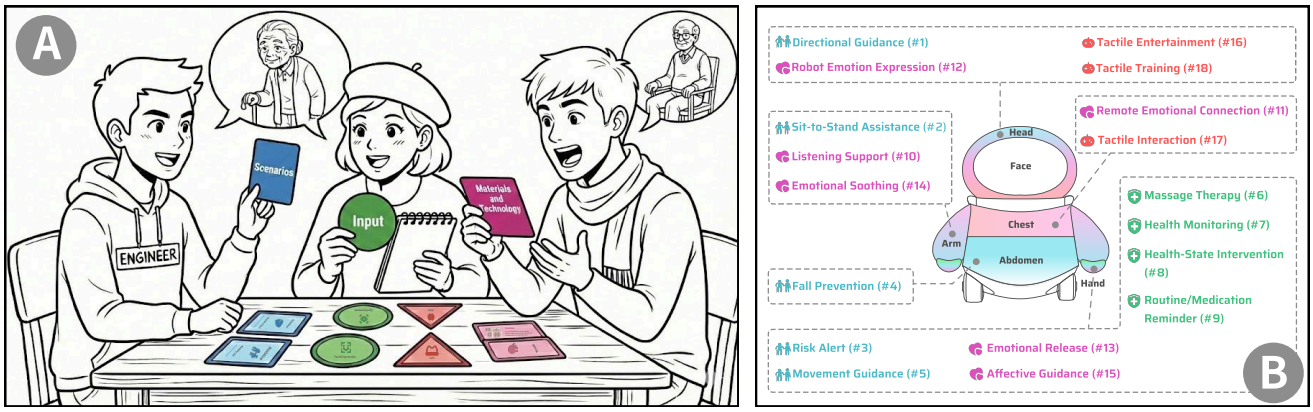

Figure 1: (A) Four co-design workshops where participants used cards and boards to explore soft shape-changing interfaces for older adult social robots. (B) A design space distilled from 18 concepts, organized by function and tactile roles of soft shape-changing interfaces and mapped to robot body areas.

Abstract

As social robots move into older adult care and companionship, soft shape-changing interfaces offer unique potential for safe and emotional tactile interaction. Yet, prior work often focuses on materials engineering, with limited design methods oriented towards older adults' wellbeing. We present co-design materials for early exploration, including a staged workflow, 48 design cards, and unified robot form boards. We conducted four workshops with 12 interdisciplinary participants acting as family caregivers and proxies. The workshops yielded 18 concepts and a rich design space categorized into Functional Assistance, Physiological Regulation, Affective Companionship, and Tactile Play. This study proposes a restrained and steady tactile paradigm to inform the design of wellbeing-oriented shape-changing social robots for older adult care.

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CCS Concepts

• **Human-centered computing → Human computer interaction (HCI)**; • **User studies**;

Keywords

social robot, older adults, wellbeing, tactile interaction, shape-changing interface

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1 Introduction

Social robots are autonomous systems with social intelligence that interact with people in socially acceptable ways [18]. Robots such as PARO [49], MiRo [17], Lovot [37] and Ropet [9] have been widely used in entertainment, companionship, education and healthcare to address users' social, emotional and physical needs. Many robots with human or animal characteristics also support the wellbeing of older adults [53, 75]. Studies show that social robots can improve sleep quality [58, 70], reduce the use of psychotropic medication among people with severe dementia [28] and increase communication willingness and engagement. By providing gentle stimulation, relaxation and comfort [31], these robots can ease loneliness [7, 58], stress and anxiety [52]and enhance overall quality of life [8, 29].

Social robots rely not only on visual and auditory cues but also on touch and posture to convey nonverbal emotions [45]. Studies show that tactile actions such as hugging [11, 63], handshaking [4, 47] and kissing [59] can strengthen emotional bonding, and soft materials provide a sense of safety, relaxation and positive affect than rigid shells [2]. Advances in flexible materials and soft robotics have enabled shape-changing tactile interfaces that express emotions through texture changes [25, 26] or simulate animal behaviours such as breathing or heartbeat [79, 80]. However, research on tactile and shape-changing interfaces for social robots mostly focuses on materials and engineering, with limited exploration of design approaches that support the wellbeing of older adults.

Although social robots show great potential, older adults often hold negative attitudes toward unfamiliar technologies [71]. Those with cognitive or communication limitations may feel stressed during design workshops and struggle to articulate their tactile preferences [65, 74]. To bridge this gap, we adopted a co-design approach [3, 43] and invited family caregivers as proxies. Living with or caring for older adults, these proxies understand the elders' daily habits and emotional triggers while being more open to novel technologies, allowing them to provide a grounded yet forward-looking perspective.

Through interdisciplinary workshops with 12 participants, this study makes three main contributions. First, we provide a design workflow and 48 design cards to support tactile ideation for interdisciplinary teams. Second, we distilled 18 representative soft shape-changing concepts with distinct functions and mapped them to specific locations on the social robot body. Third, we establish a rich design space and extract key interactive and ethical insights to guide the future development of gentle and low-intrusion tactile robots for older adults. Through this work, we aim to contribute design materials and design space insights that support HCI and HRI researchers in exploring innovative uses of soft shape-changing interfaces for social robots that promote older adult wellbeing.

2 Related Work

Social robots have become increasingly prevalent and are widely used in education, services, healthcare, research, and entertainment [39]. Sarrica et al. define social robots as embodied agents with varying levels of autonomy that can engage in social interaction with humans through communication, cooperation, and decision-making [61]. For older adults, social robots are commonly categorised into two groups. One group targets functional support [73, 76] and is mainly used for health monitoring [42, 46], daily activity reminders [36], and medication management [22]. The other group focuses on companionship and emotional support [14, 84]. These robots aim to reduce loneliness [55, 72] and strengthen emotional connections [16, 41] through simulated conversation, interactive games, or music playback [57, 60, 68]. Existing social robots exhibit diverse appearances and forms, including human-like robots such as NAO [19, 32] and Misty [1], animal-like robots such as Paro [54], Lovot [69], and JoyForAll [3] , as well as abstract robots such as Travis [10]. Prior studies indicate that older adults tend to prefer social robots with a medium size and animal-like characteristics [2]. Although prior research has extensively explored interaction design for social robots in older adult contexts, most work emphasises language, facial expression, and movement.

Social robots need to build connections with users at both cognitive and emotional levels, and their interaction modes typically include verbal and nonverbal forms [13]. Within nonverbal interaction, tactile feedback plays a distinct role compared to facial expression, gestures [56] and body language [67, 82], and is critical for strengthening social bonds and supporting emotional understanding [21]. Research on interpersonal touch shows that hugging activates neural pathways related to social bonding and emotion regulation [20], while handshakes can increase perceived trust and willingness to cooperate [62]. In the HRI field, the role of touch and care in emotional expression has been widely discussed [23], particularly in relation to conveying intimacy and building emotional connection [51]. Studies indicate that older adults prefer robots with soft feeling, movable body parts, and life-like characteristics [2]. However, most existing social robots rely on rigid shells or combine hard structures with fabric or fur coverings to suggest softness, which limits tactile experience and the expression of lifelike qualities. Recent advances in smart materials, including fluid materials [38, 40] and bio-inspired materials [50], offer new opportunities for robot interface design. For example, Patch-O integrates shape memory alloys with textile structures to enable diverse shape-changing motions [34], while SnapInflats uses inflatable structures to produce elastic and dynamic transformations [78]. Some studies have explored how haptic experience can be measured and computed to support design and evaluation [35, 83]. Despite these advances, such work is often driven by engineering or material performance considerations, with limited attention to user context, experiential needs, and design ethics.

Co-design methods have been widely used in design exploration within the HCI field. In the social robotics context, prior work has

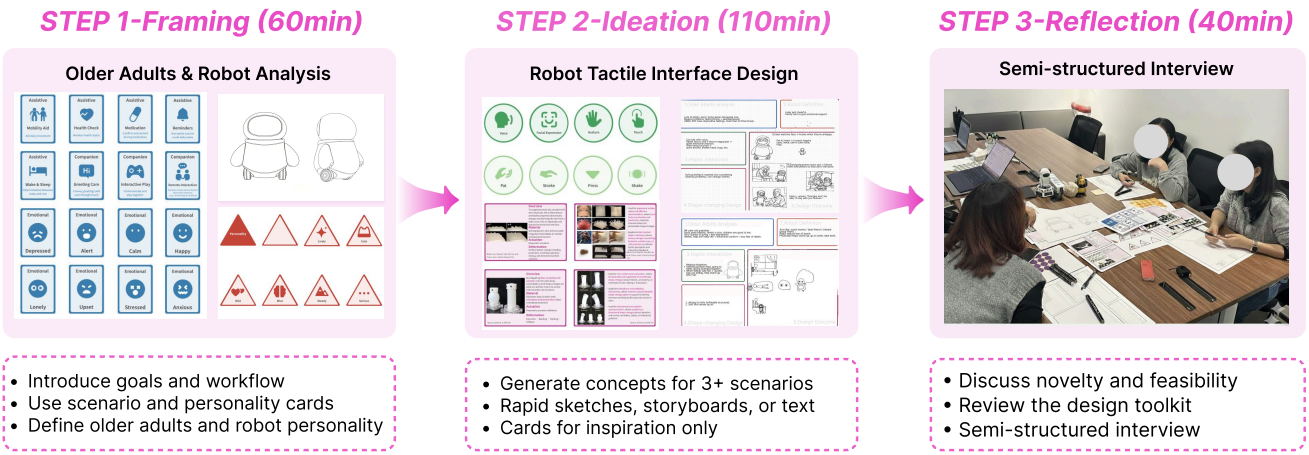

Figure 2: Workshop procedure and key activities.

involved older adults and caregivers in the design process. Stegner et al. engaged older adults and their caregivers and explored how robot-mediated interaction can support older adults' participation in assistive technology design [66]. Alhouli et al. invited younger family members to design social robots for older adults at home. Their findings indicate that a family member-centred design perspective can promote positive interactions between older adults and social robots [3]. Card-based tools have also been used to support co-design by turning research or domain knowledge into prompts for ideation and communication. Nurain et al. proposed card-based support tools to help designers understand and apply academic research findings [44]. Yu et al. developed a card-based co-design toolkit for automotive interior design to facilitate effective communication among stakeholders and explore the use of smart materials [81]. Xue et al. also used card tools in a multi-user home automation design workshop to help participants build shared understanding during ideation [77]. However, in social robotics, particularly in tactile interaction design for supporting the wellbeing of older adults, card-based co-design resources that foreground caregiver or family perspectives remain limited.

3 Methodology

3.1 Design materials

Design Workflow and Cards. Guided by the Double Diamond model [6] and the problem-solution space [5], we structured the workshop into four phases: scenario analysis, role definition, tactile interaction ideation, and soft interface envisioning. To facilitate these discussions, we initially extracted concepts from prior literature and filtered them for relevance, redundancy, and non-expert comprehensibility through three rounds of internal iteration, yielding the final 48 design cards. We also used distinct colors, shapes, and sizes to differentiate card categories and support quick lookup during the workshop. The four card categories align with the workshop phases: **(1) 16 User Scenario Cards** covering 5 older adults' activities of daily living domains [30], 3 companionship scenarios and 8 common emotional expressions [2]; **(2) 6 Personality Cards** defining robot character and interaction styles; **(3) 12 Tactile Interaction Cards** including 6 input cues, 6 output behaviours to explore input-output combinations; and **(4) 14 Shape-Changing Interface Cards** serving as technical references for embodiment. This set includes 6 metaphor cards and 8 technical cards derived from recent research (e.g., SMA [34], electro-osmotic [64], and pneumatics [15, 24, 27, 33, 48, 78]). Full contents are listed in the appendix A.

Unified Robot Form and Documentation. To minimize morphological distraction and focus on tactile interaction, we introduced a unified robot form as a shared boundary object. Its rounded upper body provides an expansive, standardized canvas for tactile mapping, while the wheeled base frames its role as a mobile companion, implying dynamic indoor and outdoor care contexts. We also used a design outcome board aligned with the four stages to document each group's reasoning path from user needs to soft shape-changing concepts, ensuring consistency across groups and supporting later analysis.

3.2 Co-Design Workshop

Participants. We recruited 12 participants across four workshop sessions via online poster announcements (9 female, 3 male; M = 31.58, SD = 8.04) as family caregiver proxies. Participants came from design (n = 5), engineering (n = 4), and sociology, nursing, or psychology (n = 3), with each session mixing disciplines to support diverse perspectives. All participants reported experience living with or caring for older adults. Robot experience and elderly care experience were self reported on a 3 point scale, where 1 indicates low familiarity and 3 indicates high familiarity. Detailed participant information is provided in Table 1.

Procedure. The 210-minute workshop followed a three-stage structure detailing in Figure 2. (1) Framing: We established a shared understanding of the design space using the scenario and personality cards. (2) Ideation: Participants worked in interdisciplinary

Table 1: Participant demographics and prior experience.

ID	Gender	Age	Educational Background	Major	Robot Experience	Older Adult Care Experience
1A	Female	19	Undergraduate	Computer Science	2	2
1B	Male	32	Undergraduate	Environmental Design	1	2
1C	Female	21	Undergraduate	Digital Media Design	1	2
2A	Female	37	Master	Fashion Design & Psychology	1	3
2B	Female	41	Undergraduate	Art History	2	3
2C	Male	38	Undergraduate	Mechanical Engineering	3	2
3A	Male	26	Master	Teleinformatics	3	2
3B	Female	32	Master	Medicine	1	3
3C	Female	23	Undergraduate	Industrial Design	1	3
4A	Female	36	Doctor	Social Science	3	3
4B	Female	30	Master	Water Conservancy	2	2
4C	Female	44	Undergraduate	Communication Design	1	3

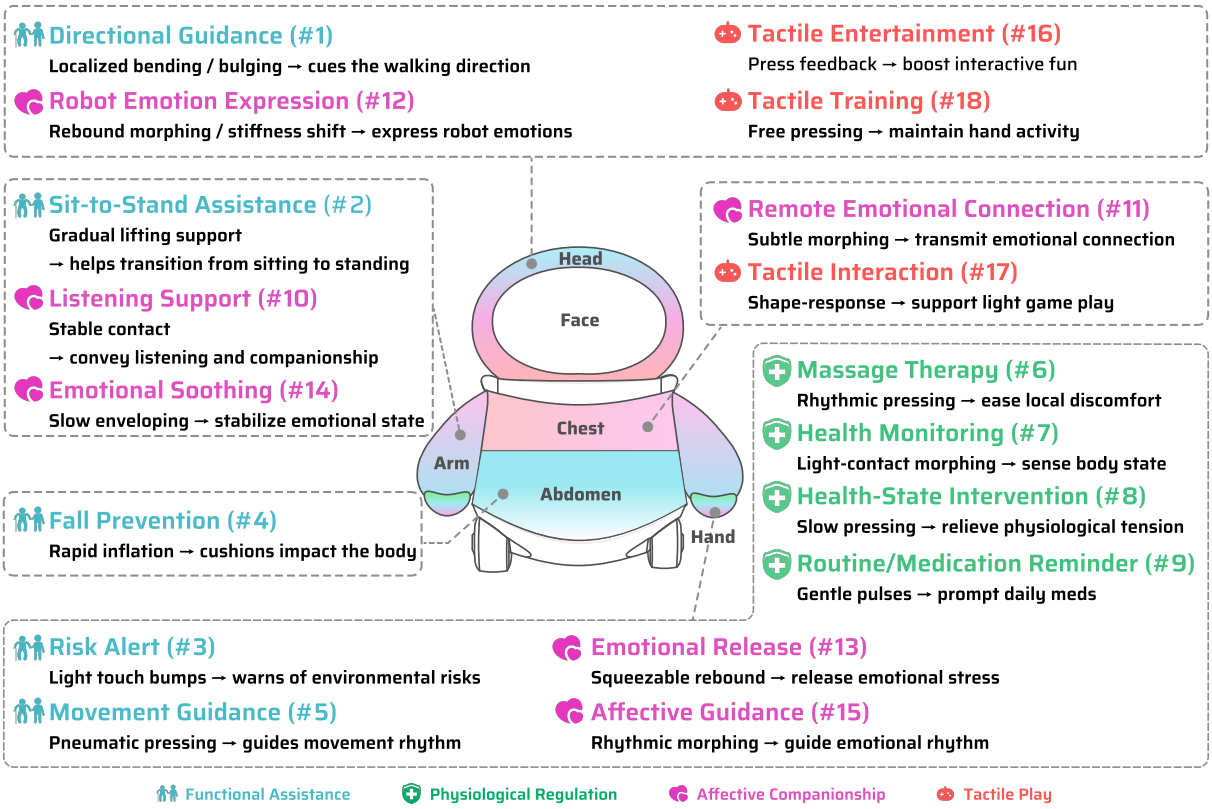

Figure 3: Design space of soft shape-changing interfaces for older adult social robots. 18 concepts are grouped into four categories and mapped to robot body areas, with each concept labeled by its ID.

groups to generate concepts, prioritizing rapid sketching over polished visuals to encourage exploration. (3) Reflection: The session concluded with a semi-structured interview, where participants discussed feasibility considerations and the efficacy of the design toolkit itself.

3.3 Data Analysis

Data analysis consisted of two parts. First, we merged functionally similar ideas to filter the 28 raw concepts down to 18 representative designs based on tactile novelty and eldercare relevance. We then classified these into 4 categories. Second, two researchers independently transcribed and coded the semi-structured interviews

using reflexive thematic analysis [12], reaching a consensus on final themes through 2 rounds of iterative discussions.

4 Results and Discussion

Spatial mapping defines a restrained and steady tactile tone. The 18 soft shape-changing concepts are grouped into Functional Assistance, Physiological Regulation, Affective Companionship, and Tactile Play. Their spatial distribution in Figure 3 suggests a clear mapping between function and proxemics, and the full set of concepts is provided in Appendix B. Hands frequently serve as low-intrusion interfaces for fine-grained interactions like health monitoring and routine reminders, where participant 2B noted that embedding sensing within natural touch can offer low-intrusion support. We emphasize that such sensing should require informed consent and easy opt-out. The torso supports high-safety needs and deep comfort through interventions like fall prevention (#4) and emotional soothing (#14), while the head transforms into a tangible interface for emotion expression and tactile play (#16). Crucially, participants agreed that gentle interaction for older adults requires behavioral restraint rather than just material softness. Participant 2B emphasized slow and steady touch that avoids interrupting daily activities, though Participant 4C noted that high-stakes scenarios like fall prevention still demand precise prediction and sufficient airbag thickness. Overall, low intensity, comfortable rhythms, and stable contact form a tactile paradigm that balances safety and comfort.

Design materials facilitate ideation. The design materials provided effective support during co-creation to guide discussions from scenario setup to tactile interface design. The material and technology cards contributed the most, as Participants 2B and 4C noted that these cards helped them recognize the simulation capabilities of current soft materials and expanded their imagination regarding tactile interface feasibility. However, purely textual and visual cards introduced interpretation costs when representing touch. Participant 1A reported that heavy text descriptions consumed cognitive resources, requiring the team to expend more effort deducing specific tactile solutions, while Participant 4B indicated that Metaphor cards presented difficulties in mapping abstract sensations to specific functions. This suggests a need to reduce the conversion cost from semantics to tactile mechanisms. For future tactile design workshops, we recommend complementing cards with hands-on material probes and richer examples, such as short videos or simple technical descriptors of tactile cues. These additions can help participants map abstract sensations to tactile experience more directly.

Interdisciplinary collaboration supports a dynamic balance between function and ethics. Participants highly praised the interdisciplinary collaboration in the workshop, noting that it directly shaped the breadth and depth of the design space. Engineering backgrounds emphasized safety baselines like dynamic obstacle avoidance and spatial interference, while medical backgrounds brought pragmatic needs such as non-intrusive monitoring for older adults with disabilities, and art and social science perspectives injected emotional warmth into the concepts. Participant 4C positively noted that this collision not only supported concept generation but also promoted knowledge exchange and broadened perspectives. However, the negotiation among disciplines regarding cost, technical feasibility, and the scale of emotional intervention also emerged. For example, regarding highly realistic soft hugging concepts like Listening support (#10), Participant 4A raised an ethical warning about handling emotional boundaries carefully to prevent older adults from developing psychological dependency caused by over-attachment. Based on this insight, we suggest moderate anthropomorphism as a core guideline, prioritizing low-intensity, predictable, and exitable tactile strategies to gradually form a tactile ethical standard in prototyping that balances technical feasibility with psychological boundaries.

Proxy perspective provides a critical anchor for a complete care ecosystem. Bringing in family members as proxies helped ground the workshop in a more realistic eldercare context. Under this perspective, the robot is not only a functional tool to relieve physical care burdens but also reshaped as a physical extension of family emotional connection. The Remote emotional connection concept (#11) reflects this direction by using shape change to recreate hugs, allowing distant children to express care with low effort. Meanwhile, proxies often focus more on safety baselines and care efficiency. This provides important realistic constraints for early concepts and suggests the need to further balance the autonomy and dignity of older adults in future validations. Some concepts lean towards low intrusion solutions like Health state intervention (#8) to reduce interruptions, which need explicit permission boundaries and user-controlled information transparency to protect elder autonomy.

5 Conclusion and Future Work

We propose soft tactile design materials that guided 12 workshop participants to co-create 18 shape-changing concepts for social robots supporting older adult wellbeing. Our findings advocate for a restrained and steady tactile paradigm and highlight the proxy perspective to balance care efficiency with elder autonomy. Future work will examine technical feasibility and implementation constraints while engaging older adults and care experts to refine care needs, interaction mechanisms, and consent and transparency requirements for gentle touch. We will then prototype representative soft shape-changing interfaces and evaluate them through controlled user studies, followed by longer-term in-home deployments to assess efficacy and ethical acceptability.

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References

[1] Giulio Antonio Abbo, Gloria Desideri, Tony Belpaeme, and Micol Spitale. 2025. "Can you be my mum?": Manipulating Social Robots in the Large Language Models Era. In *2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI)*. IEEE, 1181–1185.

[2] Saleh Alhouli et al. 2023. Older adults' emotional challenges and co-design preferences for a social robot after the COVID-19 pandemic. In *IEEE RO-MAN*. 2245–2252.

[3] Sarah Yousef Alhouli, Nora Abdullah Almania, Muneeb Ahmad, and Deepak Sahoo. 2025. Designing a Multimodal Robot Pet for Older Adults by Young Adults. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–9.

[4] Joao Avelino et al. 2018. Towards natural handshakes for social robots. *Paladyn Journal of Behavioral Robotics* 9, 1 (2018), 221–234.

[5] Minja Axelsson, Raquel Oliveira, Mattia Racca, and Ville Kyrki. 2021. Social robot co-design canvases: A participatory design framework. *ACM Transactions on Human-Robot Interaction (THRI)* 11, 1 (2021), 1–39.

[6] Jonathan Ball. 2019. *The Double Diamond: A universally accepted depiction of the design process.* https://www.designcouncil.org.uk/our-resources/archive/articles/double-diamond-universally-accepted-depiction-design-process/

[7] Marian R Banks, Linda M Willoughby, and William A Banks. 2008. Animal-assisted therapy and loneliness in nursing homes: Use of robotic vs living dogs. *Journal of the American Medical Directors Association* 9 (2008), 173–177. doi:10.1016/j.jamda.2007.11.007

[8] Roger Bemelmans, Gert Jan Gelderblom, Pieter Jonker, and Luc de Witte. 2015. Effectiveness of robot Paro in intramural psychogeriatric care. *Journal of the American Medical Directors Association* 16 (2015), 946–950. doi:10.1016/j.jamda.2015.05.007

[9] Clara Berridge, Yizhou Zhou, Julie M Robillard, and Jeffrey Kaye. 2023. Companion robots to mitigate loneliness among older adults: Perceptions of benefit and possible deception. *Frontiers in Psychology* 14 (2023), 1106633.

[10] Gurit E Birnbaum, Moran Mizrahi, Guy Hoffman, Harry T Reis, Eli J Finkel, and Omri Sass. 2016. What robots can teach us about intimacy: The reassuring effects of robot responsiveness to human disclosure. *Computers in Human Behavior* 63 (2016), 416–423.

[11] Alexander E Block et al. 2021. The six hug commandments. In *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*. 380–388.

[12] Virginia Braun and Victoria Clarke. 2022. Conceptual and design thinking for thematic analysis. *Qualitative psychology* 9, 1 (2022), 3.

[13] Cynthia Breazeal, Kerstin Dautenhahn, and Takayuki Kanda. 2016. Social robotics. *Springer handbook of robotics* (2016), 1935–1972.

[14] Shu-Chuan Chen, Wendy Moyle, Cindy Jones, and Helen Petsky. 2020. A social robot intervention on depression, loneliness, and quality of life for Taiwanese older adults in long-term care. *International psychogeriatrics* 32, 8 (2020), 981–991.

[15] Yu-Wen Chen, Wei-Ju Lin, Yi Chen, and Lung-Pan Cheng. 2021. PneuSeries: 3D shape forming with modularized serial-connected inflatables. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. 431–440.

[16] Li Chu, Hung-Wen Chen, Pei-Yi Cheng, Pokuan Ho, I-Tan Weng, Pei-Ling Yang, Sung-En Chien, Yun-Chen Tu, Chien-Chun Yang, Te-Mei Wang, et al. 2019. Identifying features that enhance older adults' acceptance of robots: a mixed methods study. *Gerontology* 65, 4 (2019), 441–450.

[17] Emily C Collins, Tony J Prescott, Ben Mitchinson, and Sebastian Conran. 2015. MIRO: a versatile biomimetic edutainment robot. In *Proceedings of the 12th International Conference on Advances in Computer Entertainment Technology*. ACM, 1–4. doi:10.1145/2832932.2832978

[18] Kerstin Dautenhahn. 2007. Socially intelligent robots: dimensions of human–robot interaction. *Philosophical Transactions of the Royal Society B: Biological Sciences* 362, 1480 (2007), 679–704.

[19] Jenny L Davis, Robert Armstrong, Anne Groggel, Sharni Doolan, Jake Sheedy, Tony P Love, and Damith Herath. 2023. Role-taking and robotic form: an exploratory study of social connection in human-robot interaction. *International Journal of Human-Computer Studies* 178 (2023), 103094.

[20] Aljoscha Dreisoerner, Nina M Junker, Wolff Schlotz, Julia Heimrich, Svenja Bloemeke, Beate Ditzen, and Rolf van Dick. 2021. Self-soothing touch and being hugged reduce cortisol responses to stress: A randomized controlled trial on stress, physical touch, and social identity. *Comprehensive Psychoneuroendocrinology* 8 (2021), 100091.

[21] Tiffany Field. 2010. Touch for socioemotional and physical well-being: A review. *Developmental review* 30, 4 (2010), 367–383.

[22] David Fischinger, Peter Einramhof, Konstantinos Papoutsakis, Walter Wohlkinger, Peter Mayer, Paul Panek, Stefan Hofmann, Tobias Koertner, Astrid Weiss, Antonis Argyros, et al. 2016. Hobbit, a care robot supporting independent living at home: First prototype and lessons learned. *Robotics and autonomous systems* 75 (2016), 60–78.

[23] Raul Hakli, Pekka Mäkelä, and Johanna Seibt. 2023. *Social Robots in Social Institutions: Proceedings of Robophilosophy 2022*. Vol. 366. IOS Press.

[24] Takayuki Hirai, Satoshi Nakamaru, Yoshihiro Kawahara, and Yasuaki Kakehi. 2018. xslate: A stiffness-controlled surface for shape-changing interfaces. In *Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–4.

[25] Yuhan Hu and Guy Hoffman. 2019. Using skin texture change to design emotion expression in social robots. In *Proceedings of the 14th ACM/IEEE International Conference on Human-Robot Interaction*. 2–10.

[26] Yuhan Hu and Guy Hoffman. 2023. What Can a Robot's Skin Be? Designing Texture-changing Skin for Human–Robot Social Interaction. *ACM Transactions on Human-Robot Interaction* 12, 2 (2023), 26.

[27] Yuhan Hu and Guy Hoffman. 2023. What can a robot's skin be? Designing texture-changing skin for human–robot social interaction. *ACM Transactions on Human-Robot Interaction* 12, 2 (2023), 1–19.

[28] Nina Jøranson, Ida Pedersen, Anne Mette Rokstad, and Camilla Ihlebaek. 2016. Change in quality of life in older people with dementia participating in Paro-activity. *Journal of Advanced Nursing* 72 (2016), 3020–3033. doi:10.1111/jan.13076

[29] Masao Kanamori, Mizue Suzuki, Hajime Oshiro, Misayo Tanaka, Tomoko Inoguchi, Hidenari Takasugi, Yoshiko Saito, and Tomoji Yokoyama. 2003. Pilot study on improvement of quality of life among elderly using a pet-type robot. In *Proceedings 2003 IEEE international symposium on computational intelligence in robotics and automation. computational intelligence in robotics and automation for the new millennium (Cat. No. 03EX694)*, Vol. 1. IEEE, 107–112.

[30] Sidney Katz, Amasa B Ford, Roland W Moskowitz, Beverly A Jackson, and Marjorie W Jaffe. 1963. Studies of illness in the aged: the index of ADL: a standardized measure of biological and psychosocial function. *jama* 185, 12 (1963), 914–919.

[31] Wada Kazuyoshi et al. 2005. Robot assisted activity at a health service facility for the aged for 17 months. In *IEEE Workshop on Advanced Robotics and its Social Impacts*. Nagoya, Japan.

[32] Ki Joon Kim, Eunil Park, and S Shyam Sundar. 2013. Caregiving role in human–robot interaction: A study of the mediating effects of perceived benefit and social presence. *Computers in Human Behavior* 29, 4 (2013), 1799–1806.

[33] Amy Koike, Michael Wehner, and Bilge Mutlu. 2024. Sprout: Designing expressivity for robots using fiber-embedded actuator. In *Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction*. 403–412.

[34] Pin-Sung Ku, Kunpeng Huang, and Cindy Hsin-Liu Kao. 2022. Patch-o: deformable woven patches for on-body actuation. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–12.

[35] Yixuan Li, Rui Zhang, Liwen He, and Yang Jiao. 2024. HapticMetric: A Smartphone Haptic Experience Computing System. In *Proceedings of the Twelfth International Symposium of Chinese CHI*. 561–570.

[36] Samantha M Loi, Alice Bennett, Marta Pearce, Khanh Nguyen, Nicola T Lautenschlager, Rajiv Khosla, and Dennis Velakoulis. 2018. A pilot study exploring staff acceptability of a socially assistive robot in a residential care facility that accommodates people under 65 years old. *International psychogeriatrics* 30, 7 (2018), 1075–1080.

[37] LOVOT. 2025. *Lovot Home Robot.* Retrieved January 21, 2026 from https://lovot.life/en/

[38] Qiuyu Lu, Jifei Ou, João Wilbert, André Haben, Haipeng Mi, and Hiroshi Ishii. 2019. milliMorph–Fluid-Driven Thin Film Shape-Change Materials for Interaction Design. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. 663–672.

[39] Hamza Mahdi, Sami Alperen Akgun, Shahed Saleh, and Kerstin Dautenhahn. 2022. A survey on the design and evolution of social robots—Past, present and future. *Robotics and Autonomous Systems* 156 (2022), 104193.

[40] Hila Mor, Tianyu Yu, Ken Nakagaki, Benjamin Harvey Miller, Yichen Jia, and Hiroshi Ishii. 2020. Venous materials: towards interactive fluidic mechanisms. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14.

[41] Wendy Moyle, Cindy Jones, Marie Cooke, Siobhan O'Dwyer, Billy Sung, and Suzie Drummond. 2014. Connecting the person with dementia and family: a feasibility study of a telepresence robot. *BMC geriatrics* 14, 1 (2014), 7.

[42] Eunkyung Na, Yoonhyuk Jung, and Seongcheol Kim. 2023. How do care service managers and workers perceive care robot adoption in elderly care facilities? *Technological Forecasting and Social Change* 187 (2023), 122250.

[43] Caterina Neef, Katharina Linden, Sophie Killmann, Julia Arndt, Nathalie Weßels, and Anja Richert. 2022. GeneRobot: How Participatory Development of Social Robots for Assisted Living Brings Generations Together. In *International Conference on Social Robotics*. Springer, 498–507.

[44] Novia Nurain, Chia-Fang Chung, Clara Caldeira, and Kay Connelly. 2024. Designing a Card-Based Design Tool to Bridge Academic Research & Design Practice For Societal Resilience. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–20.

[45] Allison Okamura. 2018. Haptic Dimensions of Human-Robot Interaction. *ACM Transactions on Human-Robot Interaction* 7, 1 (2018), 1–3.

[46] Richelle ACM Olde Keizer, Lex van Velsen, Mathieu Moncharmont, Brigitte Riche, Nadir Ammour, Susanna Del Signore, Gianluca Zia, Hermie Hermens, and Aurèle N'Dja. 2019. Using socially assistive robots for monitoring and preventing frailty among older adults: a study on usability and user experience challenges. *Health and technology* 9, 4 (2019), 595–605.

[47] Paulo Orefice et al. 2018. Pressure variation study in human-human and human-robot handshakes. In *IEEE RO-MAN*. 247–254.

[48] Jifei Ou, Mélina Skouras, Nikolaos Vlavianos, Felix Heibeck, Chin-Yi Cheng, Jannik Peters, and Hiroshi Ishii. 2016. aeroMorph-heat-sealing inflatable shape-change materials for interaction design. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology*. 121–132.

[49] PARO Robots. 2026. *PARO Therapeutic Robot.* Retrieved January 21, 2026 from http://www.parorobots.com/

[50] Pat Pataranutaporn, Angela Vujic, David S Kong, Pattie Maes, and Misha Sra. 2020. Living bits: Opportunities and challenges for integrating living microorganisms in human-computer interaction. In *Proceedings of the augmented humans international conference*. 1–12.

[51] Mark Paterson. 2023. Social robots and the futures of affective touch. *The Senses and Society* 18, 2 (2023), 110–125.

[52] Susanne Petersen, Sara Houston, Hongtu Qin, Christine Tague, and Jennifer Studley. 2017. The utilization of robotic pets in dementia care. *Journal of Alzheimer's Disease* 55 (2017), 569–574. doi:10.3233/JAD-160703

[53] Lihui Pu, Wendy Moyle, Cindy Jones, and Michael Todorovic. 2019. The effectiveness of social robots for older adults: a systematic review and meta-analysis of randomized controlled studies. *The gerontologist* 59, 1 (2019), e37–e51.

[54] Natasha Randall, Casey C Bennett, Selma Šabanović, Shinichi Nagata, Lori Eldridge, Sawyer Collins, and Jennifer A Piatt. 2019. More than just friends: in-home use and design recommendations for sensing socially assistive robots (SARs) by older adults with depression. *Paladyn* 10, One (2019).

[55] Teemu Rantanen, Paula Lehto, Pertti Vuorinen, and Kirsi Coco. 2018. The adoption of care robots in home care—A survey on the attitudes of Finnish home care personnel. *Journal of clinical nursing* 27, 9-10 (2018), 1846–1859.

[56] Niyati Rawal and Ruth Maria Stock-Homburg. 2022. Facial emotion expressions in human–robot interaction: A survey. *International Journal of Social Robotics* 14, 7 (2022), 1583–1604.

[57] Hayley Robinson, Elizabeth Broadbent, and Bruce MacDonald. 2016. Group sessions with P aro in a nursing home: Structure, observations and interviews. *Australasian journal on ageing* 35, 2 (2016), 106–112.

[58] Hannah Robinson, Bruce MacDonald, Ngaire Kerse, and Elizabeth Broadbent. 2013. The psychosocial effects of a companion robot: A randomized controlled trial. *Journal of the American Medical Directors Association* 14 (2013), 661–667. doi:10.1016/j.jamda.2013.02.007

[59] Ehsan Saadatian et al. 2014. Mediating intimacy in long-distance relationships using kiss messaging. *International Journal of Human-Computer Studies* 72, 10 (2014), 736–746.

[60] Miguel A. Salichs, Álvaro Castro-González, Elvira Salichs, Estefanía Fernández-Rodicio, Miguel Maroto-Gómez, and José J. Gamboa-Montero. 2020. Mini: a new social robot for the elderly. *International Journal of Social Robotics* 12, 6 (2020), 1231–1249.

[61] Mauro Sarrica, Sonia Brondi, and Leopoldina Fortunati. 2020. How many facets does a "social robot" have? A review of scientific and popular definitions online. *Information Technology & People* 33, 1 (2020), 1–21.

[62] Juliana Schroeder, Jane L Risen, Francesca Gino, and Michael I Norton. 2019. Handshaking promotes deal-making by signaling cooperative intent. *Journal of personality and social psychology* 116, 5 (2019), 743.

[63] Masahiro Shiomi et al. 2021. Robot reciprocation of hugs increases both interacting times and self-disclosures. *International Journal of Social Robotics* 13 (2021), 353–361.

[64] Craig Shultz and Chris Harrison. 2023. Flat panel haptics: Embedded electroosmotic pumps for scalable shape displays. In *Proceedings of the 2023 CHI conference on human factors in computing systems*. 1–16.

[65] Marijke Span et al. 2013. Involving people with dementia in the development of supportive IT applications. *Ageing Research Reviews* 12, 2 (2013), 535–551.

[66] Laura Stegner, Emmanuel Senft, and Bilge Mutlu. 2023. Situated participatory design: A method for in situ design of robotic interaction with older adults. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–15.

[67] Darja Stoeva and Margrit Gelautz. 2020. Body language in affective human-robot interaction. In *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*. 606–608.

[68] Hidenobu Sumioka, Nobuo Yamato, Masahiro Shiomi, and Hiroshi Ishiguro. 2021. A minimal design of a human infant presence: A case study toward interactive doll therapy for older adults with dementia. *Frontiers in Robotics and AI* 8 (2021), 633378.

[69] Megumi Takada, Junko Ichino, and Kaname Hayashi. 2024. A study of objective evaluation indicator based on robot activity logs for owner attachment to companion robot. *International Journal of Social Robotics* 16, 1 (2024), 125–143.

[70] Mitsuru Tanaka et al. 2012. Effect of a human-type communication robot on cognitive function in elderly women living alone. *Medical Science Monitor* 18 (2012), CR550–CR557. doi:10.12659/MSM.883350

[71] Sofia Thunberg, Maria Arnelid, and Tom Ziemke. 2022. Older Adults' Perception of the Furhat Robot. In *International Conference on Human-Agent Interaction*. 4–12.

[72] Sławomir Tobis, Mirosława Cylkowska-Nowak, Katarzyna Wieczorowska-Tobis, Mariola Pawlaczyk, and Aleksandra Suwalska. 2017. Occupational therapy students' perceptions of the role of robots in the care for older people living in the community. *Occupational therapy international* 2017, 1 (2017), 9592405.

[73] Tuuli Turja, Sakari Taipale, Marketta Niemelä, and Tomi Oinas. 2022. Positive turn in elder-care workers' views toward telecare robots. *International journal of social robotics* 14, 4 (2022), 931–944.

[74] Gubing Wang, Chiara Marradi, Armagan Albayrak, and Tischa JM van der Cammen. 2019. Co-designing with people with dementia: A scoping review of involving people with dementia in design research. *Maturitas* 127 (2019), 55–63.

[75] Sally Whelan, Kathy Murphy, Eva Barrett, Cheryl Krusche, Adam Santorelli, and Dympna Casey. 2018. Factors affecting the acceptability of social robots by older adults including people with dementia or cognitive impairment: a literature review. *International Journal of Social Robotics* 10, 5 (2018), 643–668.

[76] Ya-Huei Wu, Christine Fassert, and Anne-Sophie Rigaud. 2012. Designing robots for the elderly: appearance issue and beyond. *Archives of gerontology and geriatrics* 54, 1 (2012), 121–126.

[77] Xiao Xue, Xinyang Li, Boyang Jia, Jiachen Du, and Xinyi Fu. 2024. Who should hold control? Rethinking empowerment in home automation among cohabitants through the lens of co-design. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–19.

[78] Yue Yang, Lei Ren, Chuang Chen, Bin Hu, Zhuoyi Zhang, Xinyan Li, Yanchen Shen, Kuangqi Zhu, Junzhe Ji, Yuyang Zhang, et al. 2024. SnapInflatables: Designing Inflatables with Snap-through Instability for Responsive Interaction. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. 1–15.

[79] Steve Yohanan and Karon E MacLean. 2011. Design and assessment of the haptic creature's affect display. In *Proceedings of the 6th ACM/IEEE International Conference on Human-Robot Interaction*. 473–480.

[80] Steve Yohanan and Karon E MacLean. 2012. The role of affective touch in human-robot interaction. *International Journal of Social Robotics* 4, 2 (2012), 163–180.

[81] Tianyu Yu, Xuezhu Wang, Yao Lu, Kejin Yu, Xiwen Yao, Wenjing Deng, Zhiyu Li, Xueqing Li, Xiao Xue, Yue Yang, et al. 2025. A Card-based Co-Design Toolkit for Exploring Smart Material Applications with Multiple Stakeholders: A Case Study on Automotive Interior Design. In *Proceedings of the 2025 ACM Designing Interactive Systems Conference*. 1328–1348.

[82] Unai Zabala, Igor Rodriguez, José María Martínez-Otzeta, and Elena Lazkano. 2021. Expressing robot personality through talking body language. *Applied Sciences* 11, 10 (2021), 4639.

[83] Rui Zhang, Yixuan Li, Zihuang Wu, Yong Zhang, Jie Zhao, and Yang Jiao. 2024. SemanticTap: A Haptic Toolkit for Vibration Semantic Design of Smartphone. In *Proceedings of the 26th International Conference on Multimodal Interaction*. 264–273.

[84] Katalin Zsiga, András Tóth, Tamás Pilissy, Orsolya Péter, Zoltán Dénes, and Gábor Fazekas. 2018. Evaluation of a companion robot based on field tests with single older adults in their homes. *Assistive Technology* 30, 5 (2018), 259–266.

A Design cards

Table 2: Workshop design cards.

Card set	Card category	Items
Older adult	User scenario cards	Assistive: mobility aid, health check, medication, reminders, wake and sleep, {blank card} Companionship: greeting care, interactive play, remote interaction, {blank card} Emotional support: depressed, alert, calm, happy, lonely, upset, stressed, anxious, {blank card}
Social robot	Personality cards	lively, cute, gentle, wise, steady, serious, {blank card}
Tactile interaction design	Input cues cards	voice, eye movement, body movement, facial expression, gesture, touch, {blank card}
	Output behaviours cards	hug, handshake, pat, stroke, press, shake, {blank card}
Shape-changing interface design	Materials and technical cards	Electric heating: Patch-O, {blank card} Electroosmotic actuation: Flat Panel Haptics, {blank card} Pneumatic actuation: aeroMorph, xSlate, SnapInflatables, PneuSeries, Robot Skin, Sprout, {blank card}
	Metaphor cards	heartbeat, breath, raindrops, waves, clouds, breeze, {blank card}

B Concepts design of the workshop

Table 3: Soft shape-changing interface concepts for social robots (No. 1–9).

Category	No.	Function	Design description
Functional Assistance	1	Direction guidance	For navigation during walking, the robot adds touchable soft-deformation zones on the head and arms. Slow bending or bulging provides directional cues and helps the user keep moving in rhythm.
	2	Sit-to-stand assistance	For getting up in bed or toilet scenarios, soft support zones on the arms gradually bulge to lift and sustain support, easing the transition from sitting to standing and improving safety.
	3	Risk alert	For reduced risk awareness while walking, small deformation zones on shoulders or hands deliver alerts via light protrusions and stiffness shifts when potential hazards are detected.
	4	Fall prevention	For fall protection, a fast-deploy soft zone on the abdomen rapidly inflates when imbalance is detected, forming a cushioning surface to catch and spread impact.
	5	Movement guidance	For Tai Chi or Baduanjin practice, rhythmic pneumatic presses on hands mark key moments, offering direction and tempo cues for more consistent movements.
Physiological Regulation	6	Massage therapy	For leg soreness or discomfort, a controllable press zone in the hands performs pressing, or rolling deformations to simulate massage for relief and therapy.
	7	Health monitoring	For daily monitoring, physiological sensing is embedded in handshakes and touch. Light conforming contact on the hands supports low-intrusion state sensing and reduces resistance to traditional devices.
	8	Health-state intervention	When abnormal fluctuations (e.g., heart rate, blood pressure) are detected, the robot initiates slow, steady pressing or enveloping on hands and arms to ease tension, serving as a low-intrusion step before further support.
	9	Routine reminder	For medication or other reminders, a low-intensity zone on the hands delivers gradual prompts through rhythmic, gentle undulations, aiming to avoid interrupting ongoing activities.

Table 3 (continued). Soft shape-changing interface concepts for social robots (No. 10–18).

Category	No.	Function	Design description
Affective Companionship	10	Listening support	For storytelling and venting, stable, close-contact touch on hands, arms, and chest maintains presence through hand-holding or light hugging, reinforcing the feeling of being listened to.
	11	Remote emotional connection	When emotional expression is limited in remote calls, small deformations triggered by hand-holding or a light hug during the call make touch an extra emotional channel, strengthening connection.
	12	Robot emotion expression	For everyday emotional feedback, shape-changing "soft skin" on the head and face responds to touch strength with rebound or stiffness change, creating a tangible emotional reaction.
	13	Emotional release	For anxiety or stress, a repeatedly squeezable soft zone on the hands supports grasp–release cycles with rebound feedback, turning pressure into tactile action for emotional unloading.
	14	Emotional soothing	For loneliness or low mood, slow enveloping deformations on the arms and chest combine sustained contact and gentle squeezing to smooth emotion.
	15	Affective guidance	For anxiety or tension, low-frequency, predictable cyclic deformations output at the hands and transmitted through touch can guide breathing or help emotions gradually stabilize.
Tactile Play	16	Tactile entertainment	For daily fun, the head provides clear press–rebound feedback and can sync with sound or light responses, turning touch into immediate play.
	17	Tactile interaction	For light games and sensory activation, multiple pressable zones on arms, chest, and abdomen (e.g., small bumps like a whack-a-mole layout) respond to touch with varied deformations, forming a body-level input surface that boosts enjoyment and cognitive engagement.
	18	Tactile training	For reduced finger motor ability, pressable training areas on the head and face encourage repeated pressing and touch interaction to maintain hand activity and fine-motor practice.